RR-F-576B
April 10, 1974
SUPERSEDING
Fed. Spec. RR-F-576A
August 7, 1961

FEDERAL SPECIFICATION

FORM, CONCRETE PAVING, STEEL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers steel forms and accessories for concrete pavements and bases.

1.2 Classification.

1.2.1 Sizes. Steel forms covered by this specification shall be of one of the following sizes, as specified (see 6.2):

Size 8 - 8-inch (face depth).
Size 10 - 10-inch (face depth).
Size 12 - 12-inch (face depth).
Size 16 - 16-inch (face depth).
Size 24 - 24-inch (face depth).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

FSC 5440

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Office in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Dallas, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation-Packaging, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

A29 - General Requirements for Hot-Rolled and Cold-Finished Carbon and Alloy Steel Bars.
A36 - Standard for Structural Steel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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National Motor Freight Traffic Association, Inc. Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The forms shall be commercial products of the manufacturer's latest design, and shall meet the requirements specified herein. The forms shall be in sections 10 feet in length, and shall be furnished with stakes, stake pullers, wedges, end connections, and any accessories specified in the contract.

3.2 Standard commercial product. The concrete paving form shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product shall be included in the concrete paving form being furnished. Standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

3.2.1 Repair parts and service. Replacement, or repair parts, and service for the standard commercial product provided under this specification shall be available from the manufacturer's regular, or through commercial, parts distribution and/or service organizations.

3.3 First article. When specified (see 6.2), the contractor shall furnish a complete concrete paving form with accessories for first article inspection and approval (see 4.3.1 and 6.3).

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3.4 Material. Unless otherwise specified (see 6.2), the materials for forms and accessories shall be as specified herein. Materials not definitely specified shall be of the quality regularly used by the manufacturer in his commercially available product.

3.4.1 Steel plates and shapes. Steel for plates and other rolled shapes shall conform to ASTM A36; except that higher carbon steel shall be used for wedges, lock-joint connectors, and pins for dowel pin connectors where greater resistance to bending or peening effects of repeated hammering is required. On forms 12 inches or less in height, lock-joint connectors may be of cast-malleable iron.

3.4.2 Steel bars. Steel for bars, including stakes, shall conform to ASTM A29. Stakes shall be of high-carbon steel having a carbon content of not less than 0.45 percent and not more than 0.55 percent.

3.4.3 Steel forgings and castings. Steel for forgings and castings used for components such as stake pullers shall be in accordance with the industry standard practice for this type of equipment.

3.5 Design and construction.

3.5.1 Forms. Forms shall be of all steel construction. The plate thickness for the body of the forms, and for all major components, shall be not less than 1/4 inch in thickness. Each form shall be 10 feet in length, with a base width equal to or greater than the face depth. The top rail width of the form shall be not less than 2 inches nor greater than 2-1/4 inches. The top rail shall have an outboard turned down web depth of not less than 1 inch. Forms shall have self-alining joints, and shall be designed to withstand the lateral and vertical loads imposed by spreaders, finishing machines, or other types of form-riding equipment. One or both ends of the form shall have the base and top rail cut on a diagonal, so forms can be assembled to negotiate the inside curve of roadways on a minimum radius of 75 feet. Forms shall show no vertical deviation greater than 1/8 inch in a 10-foot length from the true plane of the top rail surface, and shall show no lateral deviation greater than 1/4 inch in a 10-foot length from the true plane of the vertical face surface. Forms shall not deflect more than 1/4 inch when subjected to the test specified in 4.4.

3.5.2 Stake pockets. Each 10-foot length of form shall have not less than three stake pockets securely riveted or welded to the face and base to act as braces. Each stake pocket on forms of 8-inch and 10-inch size shall accept one stake. Each stake pocket on forms 12-inch size and larger shall accept two stakes when three stake pockets are provided, or one stake when five stake pockets are provided. Each stake hole in the stake pocket shall be equipped with one or two steel wedges for securely

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wedging the form to the stake. Each stake pocket shall have the stake hole or holes slotted at a right angle to the long axis of the form. The slot clearance shall be adequate to permit driving a stake off-plumb without forcing the form off-line or grade.

3.5.3 Connectors, lock-joint, and dowel pin. Each form shall be provided with one lock-joint connector of not less than 1/4-inch thickness. The height of the lock-joint connector shall be the full depth of the form between the upper and lower flanges, less sliding clearance. The length of the lock-joint connector shall be sufficient to insure stability and rigidity. The lock-joint connector shall be located at one end of the form in such a manner as to be capable of connecting to the adjoining form. The lock-joint connector shall be a sliding device, designed to telescope into its own form. The lock-joint connector shall be formed or reinforced in such a manner as to permit repeated hammering or driving without distortion or breakage. For forms in sizes greater than 12 inch, dowel pin connectors may be used. The dowel pins shall be sliding devices, capable of retracting flush with the end of their forms. Not less than three dowel pins shall be provided at one end and an equal number of matching pin sockets at the opposite end of each form. The lock-joint connector and the dowel pins of the dowel pin connector shall be secured to their form in a manner which shall prevent removal or loss, and shall offer a minimum resistance in making a connection. Lock-joint and dowel pin connectors shall show no failure or permanent deformation when subjected to the test specified in 4.4.3; the lateral misalinement shall not exceed 1/4 inch between the adjoining vertical faces, and the vertical misalinement shall not exceed 1/8 inch between the adjoining top rail surfaces.

3.5.4 Stakes. Each stake shall be round and of not less than 1-inch diameter. Stakes shall have a hot forged point on one end and a flat face on the opposite end. Unless otherwise specified (see 6.2), the quantity and length of stakes to be furnished with each form shall be as shown in table I.

Table I Quantity and length of stakes, each form

	One hole pockets		Two hole pockets	
Form size	Quantity	Length (inch)	Quantity	Length (inch)
8	3	24		
10	3	24		
12	5	30	3	30
			3	18
16	5	36	3	36
			3	18
24	5	48	3	48
			3	24

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3.5.5 Stake pullers. Stake pullers shall be of all steel construction. Stake pullers shall be designed for one-man operation in pulling the longest stake required. The surfaces of the stake puller which are in contact with a stake when pulling shall be suitably hardened to resist wear. Unless otherwise specified (see 6.2), the number of stake pullers to be furnished shall be two plus one for each fifty forms.

3.5.6 Dowel bar holder assembly. When specified (see 6.2), the forms shall have an integral dowel bar holder assembly. The dowel bar holder assembly shall be of rigid all steel construction, and shall be permanently attached to the form by either riveting or welding. The design shall be in accordance with standard commercial practice. The dowel bar holder assembly and face of form shall have matching holes of the size and spacing as specified. Dowel bar holes shall be 1/16 inch greater in diameter than the dowel bar.

3.5.7 Groove forming strip. When specified (see 6.2), groove forming strips shall be furnished for attachment to the face of the forms. The attachment shall be accomplished by bolting or welding as specified. The groove forming strip shall be 10 feet in length and of sheet steel construction. Unless otherwise specified (see 6.2), the thickness and shape shall be the manufacturer's standard product for the size of form specified.

3.5.8 Wedges. Wedges for stake pockets shall be designed and constructed in a manner which shall prevent their removal or loss. Wedges shall have a thickness of not less than 5/16 inch.

3.6 Treatment and painting. All parts of the equipment shall be either cleaned and oiled, or cleaned, treated, and painted in accordance with the standard practice of the manufacturer.

3.7 Identification marking. The forms shall be marked in the following manner:

(a) Manufacturer's name or trademark.
(b) Size of form.
(c) Weight of form.
(d) Diameter of dowels required when dowel bar holder assembly is furnished.

Requirements for identification marking for military use shall be in accordance with MIL-STD-130.

3.8 Workmanship.

3.8.1 Steel fabrication. Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material

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shall be done by methods that will not cause injury to the metal. Shearing shall be done neatly and accurately. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Re-entrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. Cut surfaces shall be free from abnormal gouges or sharp projections.

3.8.2 Bolted connections. All bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided. All nuts, bolts, and screws shall be tight.

3.8.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be of an approved shape and shall be concentric with the rivet. Rivet head shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

3.8.4 Dowel bar holes. Holes for dowels shall be accurately punched or drilled and shall have the burrs removed.

3.8.5 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with a nationally recognized code. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

3.8.6 Castings. All castings shall be sound and free from patching or repair, misplaced coring, warping, and any other defect which reduces the castings ability to perform its intended function.

3.8.7 Forgings. Forgings shall be uniform in quality and condition, and shall be free from tears, cracks, seams, laps, internal ruptures, imbedded scale, segregations, or other defects which may detrimentally affect the suitability for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of inspection requirements

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specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. All concrete paving forms with accessories of the same size, offered for delivery at one time, shall be considered a lot for purposes of inspection. Random samples of the concrete paving forms with accessories shall be selected from each lot in accordance with MIL-STD-105. The inspection level shall be S-2. The Acceptable Quality Level shall be 6.5 percent defective.

4.3 Examination. Each concrete paving form with accessories selected for inspection shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy or suitability. This element of inspection shall encompass all visual examinations and dimensional measurements.

4.3.1 First article inspection. First article inspection shall be performed on one concrete paving form with accessories when a first article sample is required (see 3.3). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be a standard production item from the supplier's current inventory provided the concrete paving form with accessories meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining concrete paving forms and accessories to be furnished under the contract.

4.4 Tests. The first article and each of the production units comprising the sample selected in accordance with 4.2 shall be tested for acceptable performance and for compliance with all requirements of this specification where compliance can be determined by testing. The tests shall include, but shall not necessarily be limited to, the methods specified below. Any form or accessory failing to pass any test shall be rejected.

4.4.1 Straightness. A form shall be placed on a plane surface in a normal position. With the form in the normal position, a 10-foot straight edge shall be placed against the surface in question and measurements shall be taken to determine if excessive deviations from a true plane surface exist. Vertical deviations from a true plane surface shall be determined by measurements taken at right angles to the top rail surface of the form and the edge of the straight edge. Lateral deviations from a

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true plane surface shall be determined by measurements taken at right angles to the face surface of the form and the edge of the straight edge. Vertical and lateral deviations may be determined with an engineer's transit or level. Vertical or lateral deviations greater than that specified in 3.4 shall be cause for rejection.

4.4.2 Deflection. A form shall be placed in the normal position and supported as a simple beam with a clear span of 9 feet 6 inches. The form shall then be subjected to a concentrated static load of 5 tons applied on, or suspended from, the top rail at the midpoint of the span. While the form is under load, measurements shall be taken to determine if excessive deflection exists. The amount of deflection shall be determined by measurements taken at right angles to the top rail surface of the form and the edge of a straight edge, or the deflection may be determined with an engineer's transit or level. Deflection greater than that specified in 3.4 shall be cause for rejection.

4.4.3 Alinement and strength of connections. Three identical forms shall be placed and connected in a straight line on a prepared level subbase. The three forms shall then be staked and securely wedged in place. After the preceding conditions have been met, the center form shall have its subbase removed for its entire length to a depth of not less than 1 inch. The center form shall then be subjected to a concentrated static load of 5 tons applied on the top rail at 6 inches from one end. While the forms are under load, measurements shall be taken, at the joint nearest the load, to determine if excessive misalinement exists. Lateral misalinement shall be determined by measurements taken at right angles to the face of the center form at the connection with the adjoining form. Vertical misalinement shall be determined by measurements taken at right angles to the top rail surface of the center form at the connection with the adjoining form. Vertical and lateral alinement may be determined with an engineer's transit or level. Lateral or vertical misalinement greater than that specified in 3.4.3 shall be cause for rejection. After the measurements for misalinement have been completed and the 5-ton load removed from the form, a visual check for failure and permanent deformation shall be made on the connector that was nearest the point of load application. Any indication of failure such as sheared or elongated rivets, separation of welds, splits or breakes in the base metal, kinks or bends occurring in the connector or its form shall be cause for rejection.

4.4.4 Stake puller operational test. The stake puller shall be given a realistic operational test, consisting of pulling not less than 60 firmly imbedded stakes without any indication of malfunction or excessive wear.

4.5 Preparation for delivery inspection. The preservation, packaging, packing, and marking of the concrete paving forms with accessories shall be inspected to verify conformance to the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Unpainted and uncoated surfaces of the forms and component parts shall be coated with type P-1 preservative conforming to the applicable specification listed in, and applied in accordance with MIL-P-116.

5.1.2 Level C. The complete forms shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Stakes and stake pullers. Stakes and stake pullers shall be packed in close-fitting boxes conforming to PPP-B-621, class 2, or PPP-B-601, overseas type. Contents shall be blocked and braced to prevent movement within the boxes.

5.2.1.2 Forms. The forms shall be nested in pairs with the lock-joint connectors at the same end, and turned so that one base forms the bottom, and the other base the top of the bundle. The lock-joint connector sliding devices shall be telescoped into the forms. The nested sections shall be bundled into secured lifts of six nested pairs of forms, and strapped girthwise with 1-1/4 by 0.035-inch flat steel strapping conforming to QQ-S-781, type I, class B. A minimum of three straps shall be provided for each lift. Nominal 2- by 4-inch wood battens shall be placed beneath all straps on the top, bottom, and sides of the lift. End straps shall be placed not more than 18 inches from ends of lift, and intermediate straps shall be equally spaced. Strapping shall be stapled to the battens with zinc-coated staples.

5.2.2 Level B. The forms, stakes, and stake pullers shall be packed as specified for level A, except that boxes shall conform to PPP-B-621, class 1, or PPP-B-601, domestic type.

5.2.3 Level C. The concrete paving forms with accessories shall be packed for shipment in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

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5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers and bundles shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, shipping containers and bundles shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Concrete paving forms with accessories are used to control the form of concrete pavements and bases.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of form required (see 1.2.1).
(c) When a first article is required for inspection and approval (see 3.3, 4.3.1, and 6.3).
(d) Material, when other than specified herein (see 3.4).
(e) Quantity and length of stakes required, if other than as shown in table I (see 3.5.4).
(f) Quantity of stake pullers required, if other than as specified (see 3.5.5).
(g) When dowel bar holder assembly is required. Size and spacing of holes for dowels (see 3.5.6).
(h) When groove forming strip is required. Whether bolted or welded. When thickness and shape shall be other than as specified (see 3.5.7).
(i) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(j) If marking shall be in accordance with Fed. Std. No. 123 or MIL-STD-129 (see 5.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be a first production item, or it may be a standard production item from the contractor's current inventory as specified in 4.3.1. The first article should consist of one complete paving form with accessories. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test, and approval of the first article.

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6.4 Revision changes. Asterisks are not used in the revision to identify changes with respect to the previous issue, due to extensive changes involved.

MILITARY CUSTODIANS:

Army - CE
Navy - YD
Air Force - 84

Review activity:

DSA - CS

Preparing activity:

Navy - YD

Project No. 5440-0018

CIVIL AGENCY COORDINATING ACTIVITIES:

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